



1. Description

1.1. Project

Project Name	STRIX_V2
Board Name	custom
Generated with:	STM32CubeMX 6.18.1
Date	08/13/2026

1.2. MCU

MCU Series	STM32F4
MCU Line	STM32F411
MCU name	STM32F411CEUx
MCU Package	UFQFPN48
MCU Pin number	48

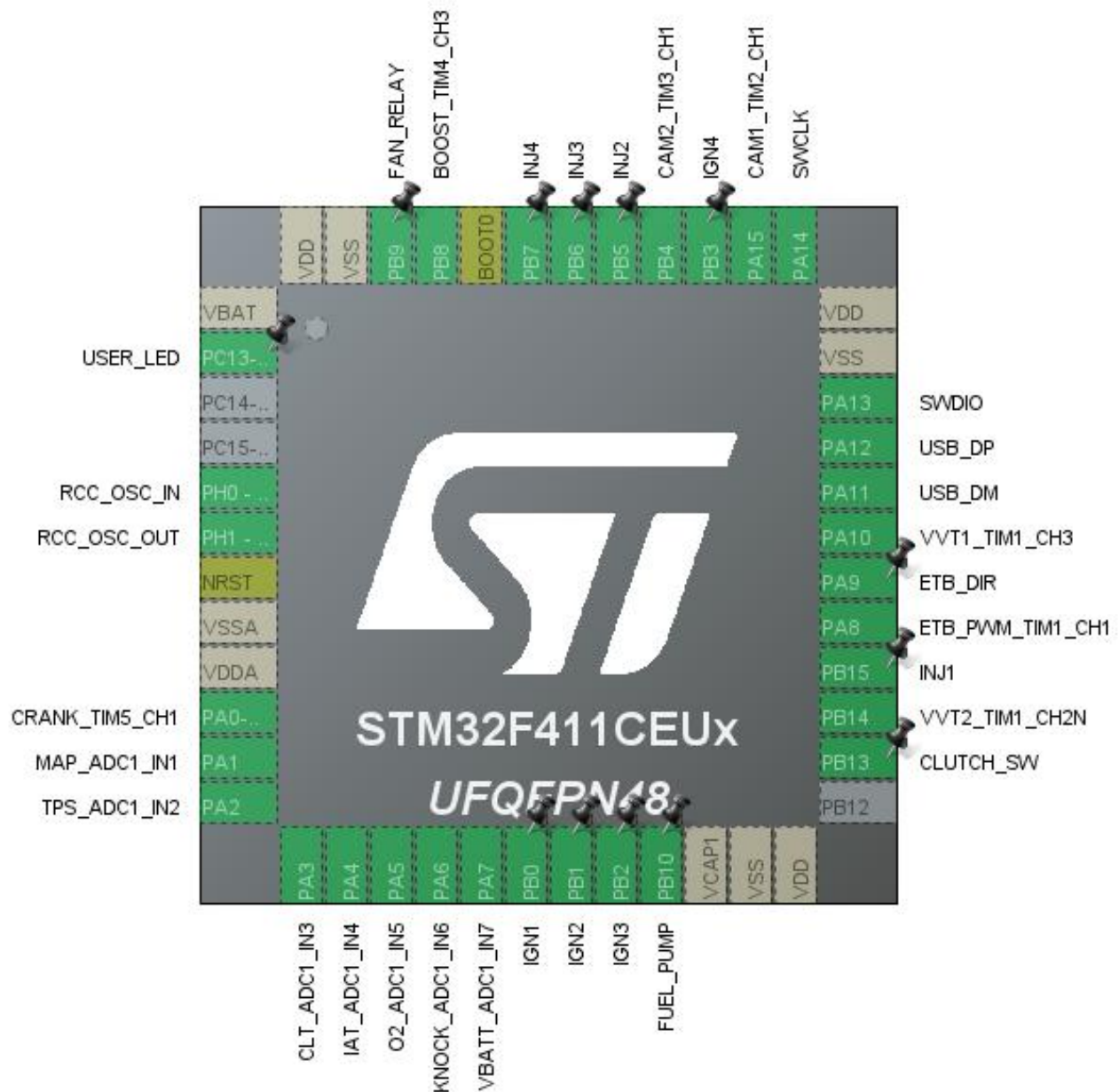
1.3. Core(s) information

Core(s)	Arm Cortex-M4
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1.4. Caution

The report was generated although the configuration was in a modified state. It may be not accurate

2. Pinout Configuration



3. Pins Configuration

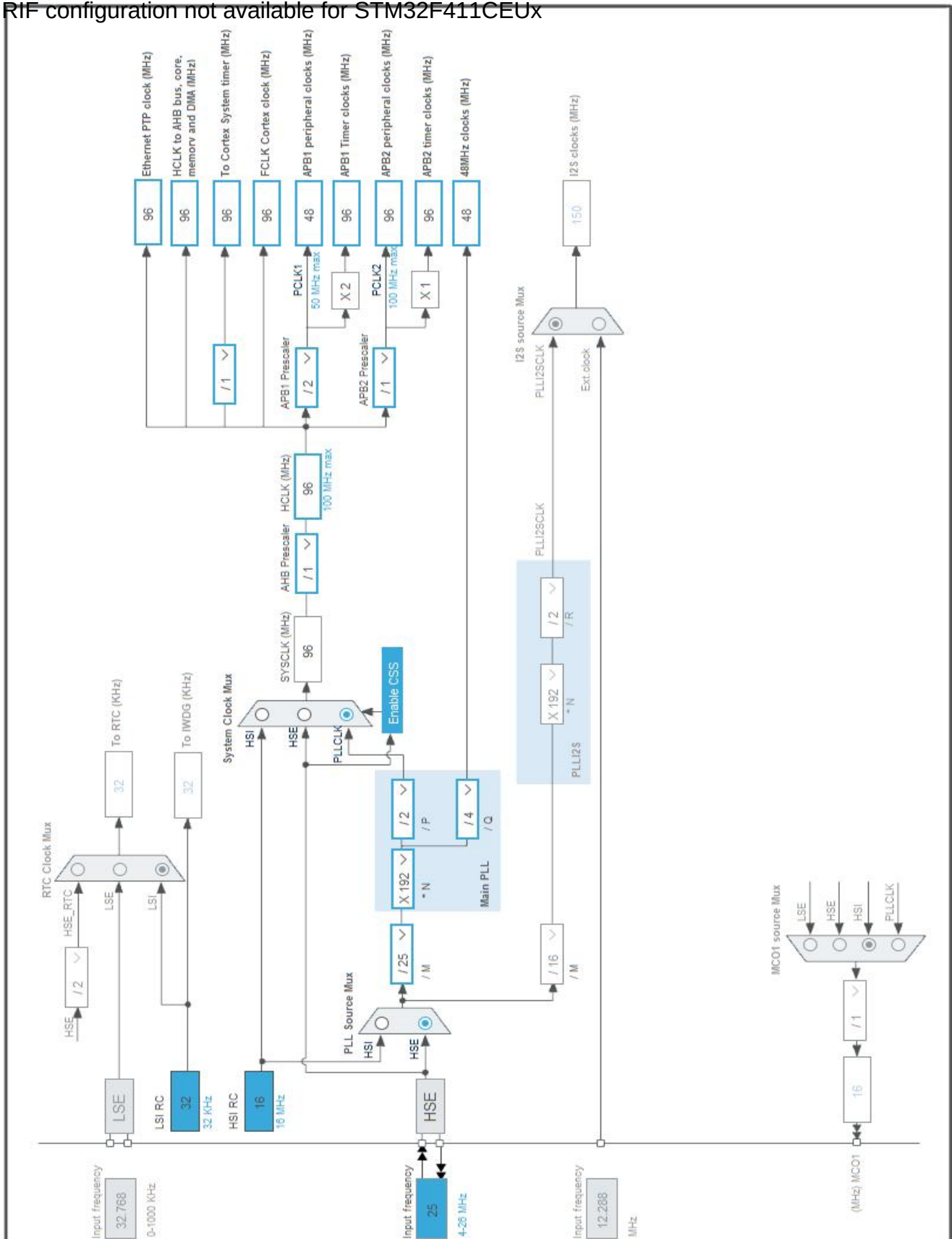
Pin Number UFQFPN48	Pin Name (function after reset)	Pin Type	Alternate Function(s)	Label
1	VBAT	Power		
2	PC13-ANTI_TAMP *	I/O	GPIO_Output	USER_LED
5	PH0 - OSC_IN	I/O	RCC_OSC_IN	
6	PH1 - OSC_OUT	I/O	RCC_OSC_OUT	
7	NRST	Reset		
8	VSSA	Power		
9	VDDA	Power		
10	PA0-WKUP	I/O	TIM5_CH1	CRANK_TIM5_CH1
11	PA1	I/O	ADC1_IN1	MAP_ADC1_IN1
12	PA2	I/O	ADC1_IN2	TPS_ADC1_IN2
13	PA3	I/O	ADC1_IN3	CLT_ADC1_IN3
14	PA4	I/O	ADC1_IN4	IAT_ADC1_IN4
15	PA5	I/O	ADC1_IN5	O2_ADC1_IN5
16	PA6	I/O	ADC1_IN6	KNOCK_ADC1_IN6
17	PA7	I/O	ADC1_IN7	VBATT_ADC1_IN7
18	PB0 *	I/O	GPIO_Output	IGN1
19	PB1 *	I/O	GPIO_Output	IGN2
20	PB2 *	I/O	GPIO_Output	IGN3
21	PB10 *	I/O	GPIO_Output	FUEL_PUMP
22	VCAP1	Power		
23	VSS	Power		
24	VDD	Power		
26	PB13	I/O	GPIO_EXTI13	CLUTCH_SW
27	PB14	I/O	TIM1_CH2N	VVT2_TIM1_CH2N
28	PB15 *	I/O	GPIO_Output	INJ1
29	PA8	I/O	TIM1_CH1	ETB_PWM_TIM1_CH1
30	PA9 *	I/O	GPIO_Output	ETB_DIR
31	PA10	I/O	TIM1_CH3	VVT1_TIM1_CH3
32	PA11	I/O	USB_OTG_FS_DM	USB_DM
33	PA12	I/O	USB_OTG_FS_DP	USB_DP
34	PA13	I/O	SYS_JTMS-SWDIO	SWDIO
35	VSS	Power		
36	VDD	Power		
37	PA14	I/O	SYS_JTCK-SWCLK	SWCLK
38	PA15	I/O	TIM2_CH1	CAM1_TIM2_CH1
39	PB3 *	I/O	GPIO_Output	IGN4

Pin Number UFQFPN48	Pin Name (function after reset)	Pin Type	Alternate Function(s)	Label
40	PB4	I/O	TIM3_CH1	CAM2_TIM3_CH1
41	PB5 *	I/O	GPIO_Output	INJ2
42	PB6 *	I/O	GPIO_Output	INJ3
43	PB7 *	I/O	GPIO_Output	INJ4
44	BOOT0	Boot		
45	PB8	I/O	TIM4_CH3	BOOST_TIM4_CH3
46	PB9 *	I/O	GPIO_Output	FAN_RELAY
47	VSS	Power		
48	VDD	Power		

* The pin is affected with an I/O function

4. Clock Tree Configuration

RIF configuration not available for STM32F411CEUX



1. Power Consumption Calculator report

1.1. Microcontroller Selection

Series	STM32F4
Line	STM32F411
MCU	STM32F411CEUx
Datasheet	DS10314 Rev6

1.2. Parameter Selection

Temperature	25
Vdd	1.7

1.3. Battery Selection

Battery	Li-SOCL2(A3400)
Capacity	3400.0 mAh
Self Discharge	0.08 %/month
Nominal Voltage	3.6 V
Max Cont Current	100.0 mA
Max Pulse Current	200.0 mA
Cells in series	1
Cells in parallel	1

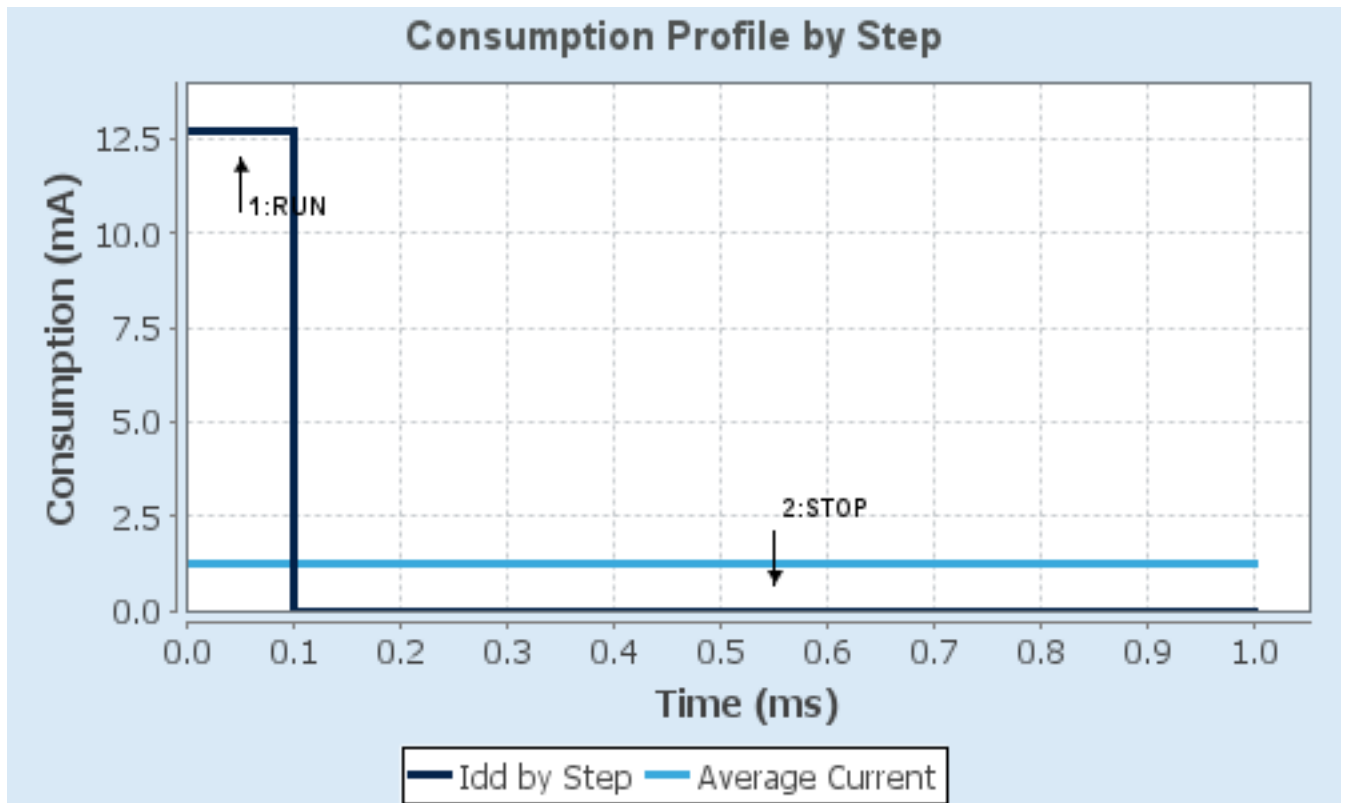
1.4. Sequence

Step	Step1	Step2
Mode	RUN	STOP
Vdd	1.7	1.7
Voltage Source	Battery	Battery
Range	Scale1-High	No Scale
Fetch Type	SRAM	n/a
CPU Frequency	100 MHz	0 Hz
Clock Configuration	HSE PLL	Regulator_LPLV Flash-PwrDwn
Clock Source Frequency	4 MHz	0 Hz
Peripherals		
Additional Cons.	0 mA	0 mA
Average Current	12.7 mA	9 μ A
Duration	0.1 ms	0.9 ms
DMIPS	125.0	0.0
Ta Max	104.31	105
Category	In DS Table	In DS Table

1.5. Results

Sequence Time	1 ms	Average Current	1.28 mA
Battery Life	3 months, 19 days, 6 hours	Average DMIPS	125.0 DMIPS

1.6. Chart



2. Software Project

2.1. Project Settings

Name	Value
Project Name	STRIX_V2
Project Folder	C:\Users\JeT\Desktop\TL_F\STRIX_V2\STRIX_V2\STRIX_V2
Toolchain / IDE	STM32CubeIDE
Firmware Package Name and Version	STM32Cube FW_F4 V1.28.0
Application Structure	Advanced
Generate Under Root	Yes
Do not generate the main()	No
Minimum Heap Size	0x200
Minimum Stack Size	0x400

2.2. Code Generation Settings

Name	Value
STM32Cube MCU packages and embedded software	Copy only the necessary library files
Generate peripheral initialization as a pair of '.c/.h' files	No
Backup previously generated files when re-generating	No
Keep User Code when re-generating	Yes
Delete previously generated files when not re-generated	Yes
Set all free pins as analog (to optimize the power consumption)	No
Enable Full Assert	No

2.3. Advanced Settings - Generated Function Calls

Rank	Function Name	Peripheral Instance Name
1	SystemClock_Config	RCC
2	MX_GPIO_Init	GPIO
3	MX_DMA_Init	DMA
4	MX_ADC1_Init	ADC1
5	MX_TIM1_Init	TIM1
6	MX_TIM2_Init	TIM2
7	MX_TIM3_Init	TIM3
8	MX_TIM4_Init	TIM4
9	MX_TIM5_Init	TIM5
10	MX_TIM9_Init	TIM9
11	MX_CRC_Init	CRC

Rank	Function Name	Peripheral Instance Name
12	MX_USB_DEVICE_Init	USB_DEVICE

3. Peripherals and Middlewares Configuration

3.1. ADC1

mode: IN1

mode: IN2

mode: IN3

mode: IN4

mode: IN5

mode: IN6

mode: IN7

3.1.1. Parameter Settings:

ADCs_Common_Settings:

Mode Independent mode

ADC_Settings:

Clock Prescaler PCLK2 divided by 4

Resolution	12 bits (15 ADC Clock cycles)
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Data Alignment Right alignment

Scan Conversion Mode Enabled

Continuous Conversion Mode **Enabled ***

Discontinuous Conversion Mode	Disabled
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DMA Continuous Requests **Enabled ***

End Of Conversion Selection **EOC flag at the end of all conversions ***

<html><img

```
src='jar:file:/C:/Users/JeT/AppData/Local/Programs/STM32CubeMX/STM32CubeMX.exe!/com/st/microxp  
lorer/image/snk/error.png' <font color=#e6007e><b> &nbsp;
```

ADC_Regular_ConversionMode:

Number Of Conversion 8 *

External Trigger Conversion Source	Regular Conversion launched by software
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External Trigger Conversion Edge	None
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Rank 1

Channel	Channel 1
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Sampling Time **84 Cycles ***

Rank 2 *

Channel	Channel 2 *
---------	-------------

Sampling Time **84 Cycles ***

Rank 3*

Channel	Channel 3 *
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Sampling Time

	480 Cycles *
<u>Rank</u>	4 *
Channel	Channel 4 *
Sampling Time	480 Cycles *
<u>Rank</u>	5 *
Channel	Channel 5 *
Sampling Time	84 Cycles *
<u>Rank</u>	6 *
Channel	Channel 6 *
Sampling Time	84 Cycles *
<u>Rank</u>	7 *
Channel	Channel 7 *
Sampling Time	144 Cycles *
<u>Rank</u>	8 *
Channel	Channel 12 *
Sampling Time	84 Cycles *

ADC_Injected_ConversionMode:

Number Of Conversions 0

WatchDog:

Enable Analog WatchDog Mode false

3.2. CRC

mode: Activated

3.3. RCC

High Speed Clock (HSE): Crystal/Ceramic Resonator

3.3.1. Parameter Settings:

System Parameters:

VDD voltage (V)	3.3
Instruction Cache	Enabled
Prefetch Buffer	Enabled
Data Cache	Enabled
Flash Latency(WS)	3 WS (4 CPU cycle)

RCC Parameters:

HSI Calibration Value	16
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TIM Prescaler Selection	Disabled
HSE Startup Timeout Value (ms)	100
LSE Startup Timeout Value (ms)	5000
Power Parameters:	
Power Regulator Voltage Scale	Power Regulator Voltage Scale 1

3.4. SYS

Debug: Serial Wire

Timebase Source: SysTick

3.5. TIM1

Clock Source : Internal Clock

Channel1: PWM Generation CH1

Channel2: PWM Generation CH2N

Channel3: PWM Generation CH3

3.5.1. Parameter Settings:

Counter Settings:

Prescaler (PSC - 16 bits value)	95 *
Counter Mode	Up
Counter Period (AutoReload Register - 16 bits value)	999 *
Internal Clock Division (CKD)	No Division
Repetition Counter (RCR - 8 bits value)	0
auto-reload preload	Disable

Trigger Output (TRGO) Parameters:

Master/Slave Mode (MSM bit)	Disable (Trigger input effect not delayed)
Trigger Event Selection	Reset (UG bit from TIMx_EGR)

Break And Dead Time management - BRK Configuration:

BRK State	Disable
BRK Polarity	High

Break And Dead Time management - Output Configuration:

Automatic Output State	Disable
Off State Selection for Run Mode (OSSR)	Disable
Off State Selection for Idle Mode (OSSI)	Disable
Lock Configuration	Off

PWM Generation Channel 1:

Mode	PWM mode 1
Pulse (16 bits value)	0

Output compare preload	Enable
Fast Mode	Disable
CH Polarity	High
CH Idle State	Reset

PWM Generation Channel 2N:

Mode	PWM mode 1
Pulse (16 bits value)	0
Output compare preload	Enable
Fast Mode	Disable
CHN Polarity	High
CHN Idle State	Reset

PWM Generation Channel 3:

Mode	PWM mode 1
Pulse (16 bits value)	0
Output compare preload	Enable
Fast Mode	Disable
CH Polarity	High
CH Idle State	Reset

3.6. TIM2

Clock Source : Internal Clock

Channel1: Input Capture direct mode

3.6.1. Parameter Settings:

Counter Settings:

Prescaler (PSC - 16 bits value)	95 *
Counter Mode	Up
Counter Period (AutoReload Register - 32 bits value)	65535 *
Internal Clock Division (CKD)	No Division
auto-reload preload	Disable

Trigger Output (TRGO) Parameters:

Master/Slave Mode (MSM bit)	Disable (Trigger input effect not delayed)
Trigger Event Selection	Reset (UG bit from TIMx_EGR)

Input Capture Channel 1:

Polarity Selection	Rising Edge
IC Selection	Direct
Prescaler Division Ratio	No division
Input Filter (4 bits value)	0

3.7. TIM3

mode: Clock Source

Channel1: Input Capture direct mode

3.7.1. Parameter Settings:

Counter Settings:

Prescaler (PSC - 16 bits value)	95 *
Counter Mode	Up
Counter Period (AutoReload Register - 16 bits value)	65535
Internal Clock Division (CKD)	No Division
auto-reload preload	Disable

Trigger Output (TRGO) Parameters:

Master/Slave Mode (MSM bit)	Disable (Trigger input effect not delayed)
Trigger Event Selection	Reset (UG bit from TIMx_EGR)

Input Capture Channel 1:

Polarity Selection	Rising Edge
IC Selection	Direct
Prescaler Division Ratio	No division
Input Filter (4 bits value)	0

3.8. TIM4

mode: Clock Source

Channel3: PWM Generation CH3

3.8.1. Parameter Settings:

Counter Settings:

Prescaler (PSC - 16 bits value)	95 *
Counter Mode	Up
Counter Period (AutoReload Register - 16 bits value)	999 *
Internal Clock Division (CKD)	No Division
auto-reload preload	Disable

Trigger Output (TRGO) Parameters:

Master/Slave Mode (MSM bit)	Disable (Trigger input effect not delayed)
Trigger Event Selection	Reset (UG bit from TIMx_EGR)

PWM Generation Channel 3:

Mode	PWM mode 1
Pulse (16 bits value)	0
Output compare preload	Enable
Fast Mode	Disable
CH Polarity	High

3.9. TIM5

mode: Clock Source

Channel1: Input Capture direct mode

3.9.1. Parameter Settings:

Counter Settings:

Prescaler (PSC - 16 bits value)	95 *
Counter Mode	Up
Counter Period (AutoReload Register - 32 bits value)	4294967295
Internal Clock Division (CKD)	No Division
auto-reload preload	Disable

Trigger Output (TRGO) Parameters:

Master/Slave Mode (MSM bit)	Disable (Trigger input effect not delayed)
Trigger Event Selection	Reset (UG bit from TIMx_EGR)

Input Capture Channel 1:

Polarity Selection	Rising Edge
IC Selection	Direct
Prescaler Division Ratio	No division
Input Filter (4 bits value)	0

3.10. TIM9

mode: Clock Source

3.10.1. Parameter Settings:

Counter Settings:

Prescaler (PSC - 16 bits value)	95 *
Counter Mode	Up
Counter Period (AutoReload Register - 16 bits value)	999 *
Internal Clock Division (CKD)	No Division
auto-reload preload	Disable

3.11. USB_OTG_FS

Mode: Device_Only

3.11.1. Parameter Settings:

Speed	Device Full Speed 12MBit/s
Low power	Disabled
Link Power Management	Disabled
VBUS sensing	Disabled
Signal start of frame	Disabled

3.12. USB_DEVICE

Class For FS IP: Communication Device Class (Virtual Port Com)

3.12.1. Parameter Settings:

Basic Parameters:

USBD_MAX_NUM_INTERFACES (Maximum number of supported interfaces)	1
USBD_MAX_NUM_CONFIGURATION (Maximum number of supported configuration)	1
USBD_MAX_STR_DESC_SIZ (Maximum size for the string descriptors)	512
USBD_SELF_POWERED (Enabled self power)	Enabled
USBD_DEBUG_LEVEL (USBD Debug Level)	0: No debug message

Class Parameters:

USB CDC Rx Buffer Size	2048
USB CDC Tx Buffer Size	2048

3.12.2. Device Descriptor:

Device Descriptor:

VID (Vendor Identifier)	1155
LANGID_STRING (Language Identifier)	English(United States)
MANUFACTURER_STRING (Manufacturer Identifier)	TorquEFI *

Device Descriptor FS:

PID (Product Identifier)	22336
PRODUCT_STRING (Product Identifier)	STRIX V2 ECU *
CONFIGURATION_STRING (Configuration Identifier)	CDC Config
INTERFACE_STRING (Interface Identifier)	CDC Interface

*** User modified value**

4. System Configuration

4.1. GPIO configuration

IP	Pin	Signal	GPIO mode	GPIO pull/up pull down	Max Speed	User Label
ADC1	PA1	ADC1_IN1	Analog mode	No pull-up and no pull-down	n/a	MAP_ADC1_IN1
	PA2	ADC1_IN2	Analog mode	No pull-up and no pull-down	n/a	TPS_ADC1_IN2
	PA3	ADC1_IN3	Analog mode	No pull-up and no pull-down	n/a	CLT_ADC1_IN3
	PA4	ADC1_IN4	Analog mode	No pull-up and no pull-down	n/a	IAT_ADC1_IN4
	PA5	ADC1_IN5	Analog mode	No pull-up and no pull-down	n/a	O2_ADC1_IN5
	PA6	ADC1_IN6	Analog mode	No pull-up and no pull-down	n/a	KNOCK_ADC1_IN6
	PA7	ADC1_IN7	Analog mode	No pull-up and no pull-down	n/a	VBATT_ADC1_IN7
RCC	PH0 - OSC_IN	RCC_OSC_IN	n/a	n/a	n/a	
	PH1 - OSC_OUT	RCC_OSC_OUT	n/a	n/a	n/a	
SYS	PA13	SYS_JTMS-SWDIO	n/a	n/a	n/a	SWDIO
	PA14	SYS_JTCK-SWCLK	n/a	n/a	n/a	SWCLK
TIM1	PB14	TIM1_CH2N	Alternate Function Push Pull	No pull-up and no pull-down	Low	VVT2_TIM1_CH2N
	PA8	TIM1_CH1	Alternate Function Push Pull	No pull-up and no pull-down	Low	ETB_PWM_TIM1_CH1
	PA10	TIM1_CH3	Alternate Function Push Pull	No pull-up and no pull-down	Low	VVT1_TIM1_CH3
TIM2	PA15	TIM2_CH1	Alternate Function Push Pull	No pull-up and no pull-down	Low	CAM1_TIM2_CH1
TIM3	PB4	TIM3_CH1	Alternate Function Push Pull	No pull-up and no pull-down	Low	CAM2_TIM3_CH1
TIM4	PB8	TIM4_CH3	Alternate Function Push Pull	No pull-up and no pull-down	Low	BOOST_TIM4_CH3
TIM5	PA0-WKUP	TIM5_CH1	Alternate Function Push Pull	No pull-up and no pull-down	Low	CRANK_TIM5_CH1
USB_OTG_FS	PA11	USB_OTG_FS_DM	Alternate Function Push Pull	No pull-up and no pull-down	Very High *	USB_DM
	PA12	USB_OTG_FS_DP	Alternate Function Push Pull	No pull-up and no pull-down	Very High *	USB_DP
GPIO	PC13-ANTI_TAMP	GPIO_Output	Output Push Pull	No pull-up and no pull-down	Low	USER_LED
	PB0	GPIO_Output	Output Push Pull	No pull-up and no pull-down	Low	IGN1
	PB1	GPIO_Output	Output Push Pull	No pull-up and no pull-down	Low	IGN2
	PB2	GPIO_Output	Output Push Pull	No pull-up and no pull-down	Low	IGN3
	PB10	GPIO_Output	Output Push Pull	No pull-up and no pull-down	Low	FUEL_PUMP
	PB13	GPIO_EXTI13	External Interrupt Mode with Rising edge trigger detection	Pull-up *	n/a	CLUTCH_SW
	PB15	GPIO_Output	Output Push Pull	No pull-up and no pull-down	Low	INJ1
	PA9	GPIO_Output	Output Push Pull	No pull-up and no pull-down	Low	ETB_DIR
	PB3	GPIO_Output	Output Push Pull	No pull-up and no pull-down	Low	IGN4

IP	Pin	Signal	GPIO mode	GPIO pull/up pull down	Max Speed	User Label
	PB5	GPIO_Output	Output Push Pull	No pull-up and no pull-down	Low	INJ2
	PB6	GPIO_Output	Output Push Pull	No pull-up and no pull-down	Low	INJ3
	PB7	GPIO_Output	Output Push Pull	No pull-up and no pull-down	Low	INJ4
	PB9	GPIO_Output	Output Push Pull	No pull-up and no pull-down	Low	FAN_RELAY

4.2. DMA configuration

DMA request	Stream	Direction	Priority
ADC1	DMA2_Stream0	Peripheral To Memory	High *

ADC1: DMA2_Stream0 DMA request Settings:

Mode: **Circular ***
Use fifo: Disable
Peripheral Increment: Disable
Memory Increment: **Enable ***
Peripheral Data Width: Half Word
Memory Data Width: Half Word

4.3. NVIC configuration

4.3.1. NVIC

Interrupt Table	Enable	Preenmption Priority	SubPriority
Non maskable interrupt	true	0	0
Hard fault interrupt	true	0	0
Memory management fault	true	0	0
Pre-fetch fault, memory access fault	true	0	0
Undefined instruction or illegal state	true	0	0
System service call via SWI instruction	true	0	0
Debug monitor	true	0	0
Pendable request for system service	true	0	0
System tick timer	true	15	0
TIM2 global interrupt	true	1	0
TIM3 global interrupt	true	1	0
TIM5 global interrupt	true	1	0
DMA2 stream0 global interrupt	true	0	0
USB On The Go FS global interrupt	true	5	0
PVD interrupt through EXTI line 16	unused		
Flash global interrupt	unused		
RCC global interrupt	unused		
ADC1 global interrupt	unused		
TIM1 break interrupt and TIM9 global interrupt	unused		
TIM1 update interrupt and TIM10 global interrupt	unused		
TIM1 trigger and commutation interrupts and TIM11 global interrupt	unused		
TIM1 capture compare interrupt	unused		
TIM4 global interrupt	unused		
EXTI line[15:10] interrupts	unused		
FPU global interrupt	unused		

4.3.2. NVIC Code generation

Enabled interrupt Table	Select for init sequence ordering	Generate IRQ handler	Call HAL handler
Non maskable interrupt	false	true	false
Hard fault interrupt	false	true	false
Memory management fault	false	true	false
Pre-fetch fault, memory access fault	false	true	false
Undefined instruction or illegal state	false	true	false
System service call via SWI instruction	false	true	false

Enabled interrupt Table	Select for init sequence ordering	Generate IRQ handler	Call HAL handler
Debug monitor	false	false	false
Pendable request for system service	false	false	false
System tick timer	false	true	true
TIM2 global interrupt	false	true	true
TIM3 global interrupt	false	true	true
TIM5 global interrupt	false	true	true
DMA2 stream0 global interrupt	false	true	true
USB On The Go FS global interrupt	false	true	true

*** User modified value**

5. System Views

5.1. Category view

5.1.1. Current

Middleware

USB_DEVICE ✓

System Core

DMA ✓

GPIO ✓

NVIC ✓

RCC ✓

SYS ✓

Analog

ADC1 ✗

Timers

TIM1 ✓

TIM2 ✓

TIM3 ✓

TIM4 ✓

TIM5 ✓

TIM9 ✓

Connectivity

USB_FS ✓

Multimedia

Computing

CRC ✓

6. Docs & Resources

Type	Link
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